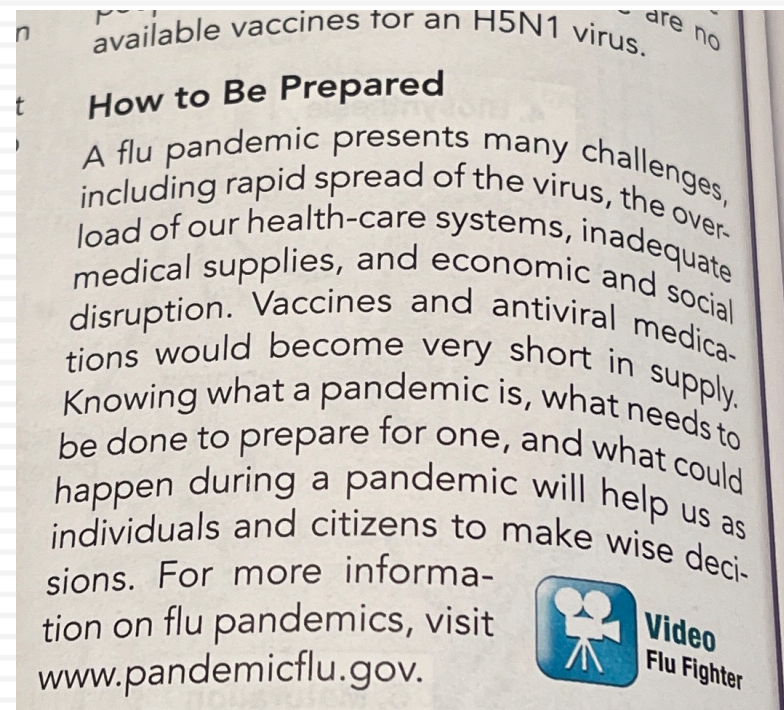


UNIT 6: DIVERSITY OF LIFE FORMS

6.1 – VIRUSES

Viruses Outline

- EQ- Does tech help to clarify or confuse classification of diversity?
- Basic Info and Classification
- Viral Structure
- Life Cycles
- Additional Info



Virus Overview

- Viruses related to a variety of plant, animal, and human disease.
- Non-cellular construction
- May have DNA or RNA genome
- Able to evolve as a result of mutations
- Require the machinery of the host cell they infect to replicate
- Not considered living
 - Do not have metabolism, cannot respond to stimuli

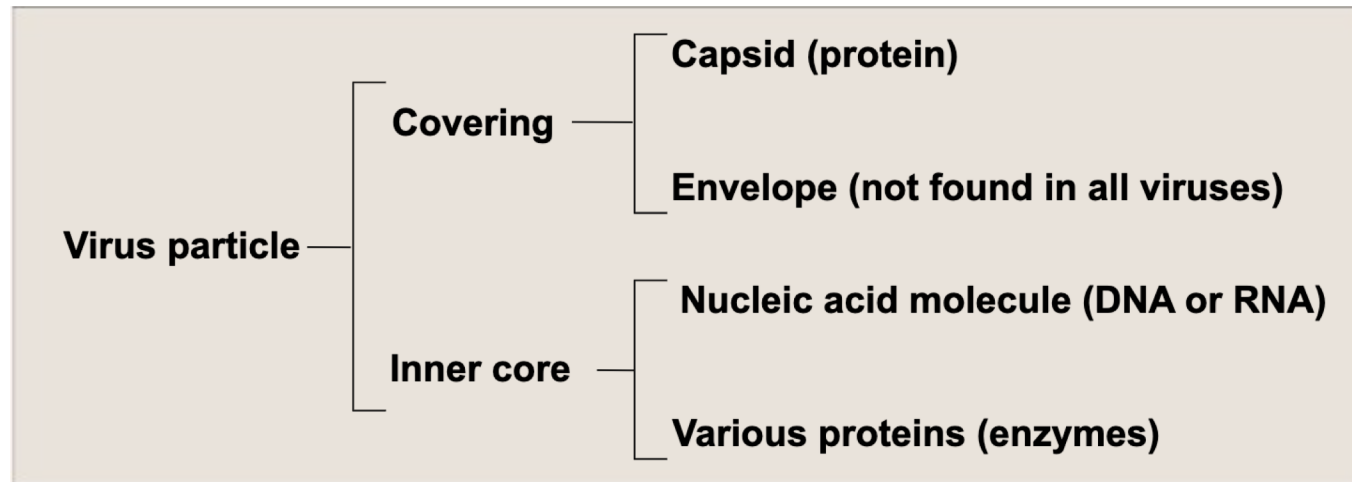
Category	Disease
Sexually transmitted diseases	AIDS (HIV), genital warts, genital herpes
Childhood diseases	Mumps, measles, chickenpox, German measles
Respiratory diseases	Common cold, influenza, severe acute respiratory syndrome (SARS)
Skin diseases	Warts, fever blisters, shingles
Digestive tract diseases	Gastroenteritis, diarrhea
Nervous system diseases	Poliomyelitis, rabies, encephalitis
Other diseases	Smallpox, hemorrhagic fevers, cancer, hepatitis, mononucleosis, yellow fever, dengue fever, conjunctivitis, hepatitis C

Virus Discovery & Classification

- Louis Pasteur – one of the first to determine objects smaller than bacteria with infectious properties
- Confirmed in mid 20th century with electron microscopy
- Classification in 2 parts
 - First sorts virus into Groups I – IV
 - Based on structure of genome (DNA vs RNA, Single vs Double Stranded)
 - Second similar to Linnaean classification
 - Order, Family, Genus, Species – No higher levels
 - Ex – Chickenpox = *Varicellovirus Zoster*
 - Seasonal flu is example of a new strain developing within a single species annually

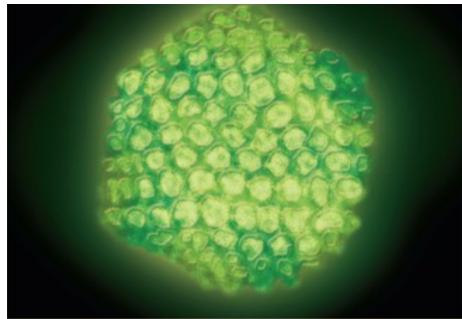
Virus Structure

- Range from 10-400nm in size
- Categorized by 1) shape, 2) genetic material, & 3) presence /absence of outer envelope



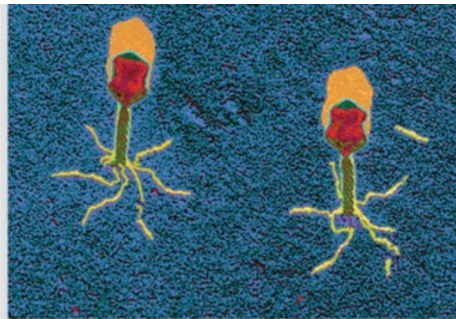
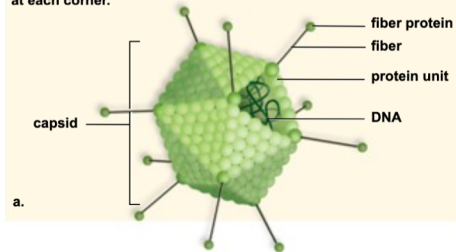
- All Viruses contain same basic anatomy otherwise:
 - Outer capsid made of protein complexes
 - Inner core of nucleic acid – 3 to 100's of genes.
- Viruses with no envelope said to be "Naked"
- Enveloped viruses covered in outer coating made of cell plasma membrane.

Virus Structure



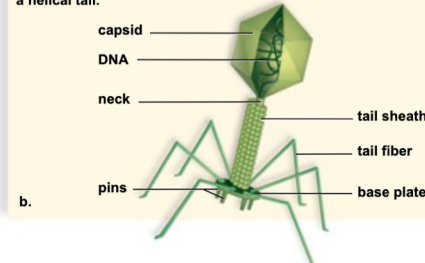
TEM 80,000X

Adenovirus: DNA virus with a polyhedral capsid and a fiber at each corner.

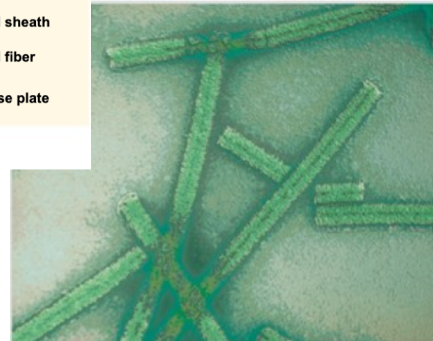


TEM 90,000X

T-even bacteriophage: DNA virus with a polyhedral head and a helical tail.

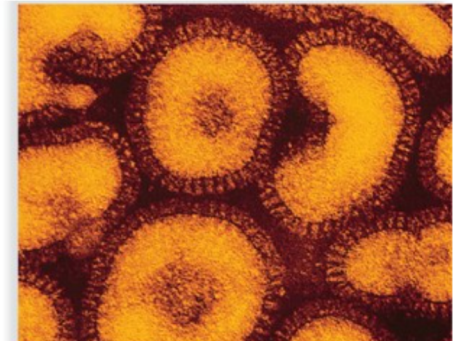
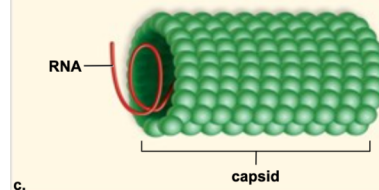


Notice common components, and differences between different types of virus. Notice the envelope structure around the last example.



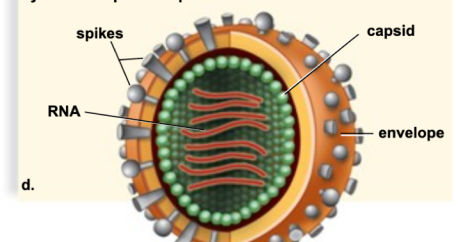
TEM 500,000X

Tobacco mosaic virus: RNA virus with a helical capsid.



20nm

Influenza virus: RNA virus with a spherical capsid surrounded by an envelope with spikes.

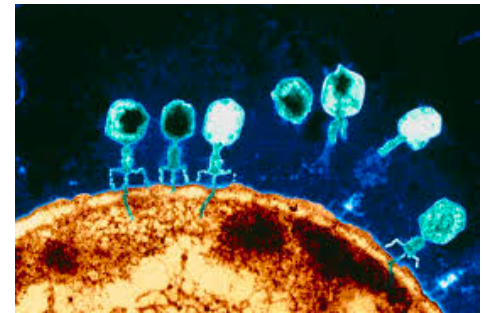


Virus Reproduction

- Technically intracellular parasites – Do not contain their own machinery for reproduction
- Gain entry into specific host cell
 - ▣ Capsid (or spikes of the envelope) adhere to specific receptor sites on the host cell surface.
- Viral nucleic acid then enters a cell
 - ▣ Viral genome codes for production of protein units in the capsid. Host cell makes the subunits to create new virus particles
- Relies on host cell enzymes, ribosomes, transfer RNA (tRNA), and ATP for its own replication
- Mistakes made in reproduction allow for mutations, and possible viral evolution as a result.

Virus Reproduction

- Bacteriophage (“phage”) – Viruses that infect bacterial cells



- Two types of viral life cycles

- Lytic Cycle

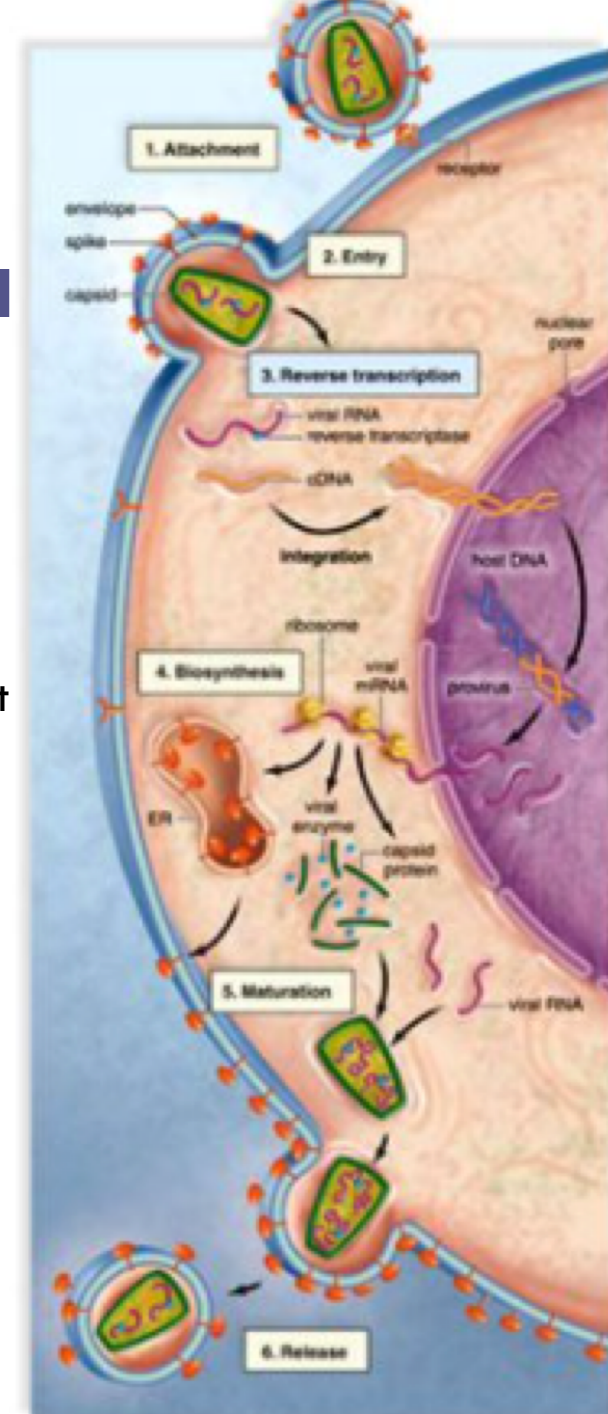
- Virus infects cell, takes over host machinery, bursts cell open to release virus
 - Attachment, Penetration, Biosynthesis, Maturation, Release
 - Ebola Virus

- Lysogenic cycle

- Virus infects cell, but may incorporate its gen. material into host.
 - Integration occurs after the penetration step
 - No immediate proliferation; can occur at later time, even years later – may enter lytic cycle at this time
 - HIV virus is lysogenic – lays dormant until a later time

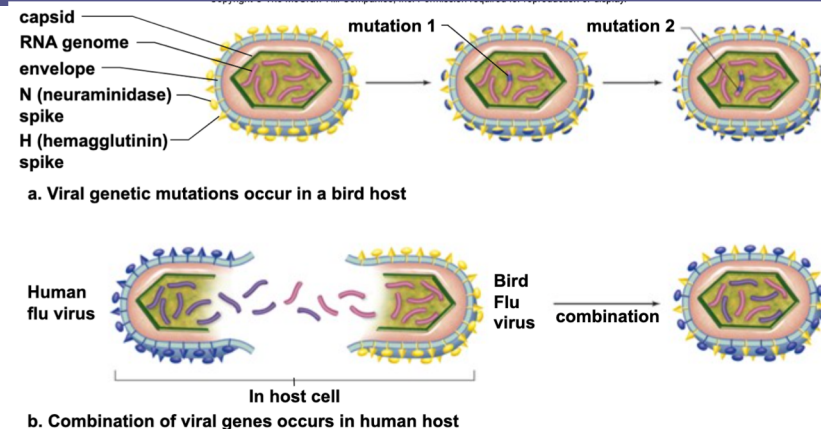
Virus Reproduction

- Animal Viruses enter host cell
- Uncoating releases viral DNA or RNA
 - ▣ If “naked virus”, release is usually lytic bursting of host cell
 - ▣ If enveloped, “budding” will occur
 - Viral particles reach cell membrane, bud forms out of the cell, and virus becomes enveloped in cell membrane coated in protein and carbohydrates.
- Retroviruses – RNA Viruses
 - ▣ Contain a special enzyme - **Reverse Transcriptase**
 - ▣ Carries out RNA → cDNA reverse transcription
 - ▣ cDNA becomes integrated into host DNA
 - Replicated as host DNA replicates
 - ▣ Viral DNA is transcribed; new viruses are produced
 - HIV, Human T Lymphotropic Virus 1



Emerging Viruses

- Viruses who have made a move between species, or infect otherwise unaffected populations
 - ▣ Ex. – H1N1 flu, West Nile, COVID-19
- Ex- Flu Pandemic
 - ▣ A flu virus has an H (hemagglutinin) spike and an N (neuraminidase) spike
 - H spike allows the virus to bind to the receptor
 - 16 different types
 - N spike attacks host plasma membranes
 - Allows mature viruses to exit the cell
 - 9 different types
 - ▣ Each type of spike can occur in different varieties
 - ▣ Our immune system only recognizes H spikes and N spikes it has been exposed to



available vaccines for an H5N1 virus. are no

How to Be Prepared

A flu pandemic presents many challenges, including rapid spread of the virus, the overload of our health-care systems, inadequate medical supplies, and economic and social disruption. Vaccines and antiviral medications would become very short in supply. Knowing what a pandemic is, what needs to be done to prepare for one, and what could happen during a pandemic will help us as individuals and citizens to make wise decisions. For more information on flu pandemics, visit www.pandemicflu.gov.

 **Video Flu Fighter**

Questions to Consider

Virioids and Prions

□ Virioids

- ▣ Naked strands of RNA – no capsid
- ▣ Associated with many crop disease
- ▣ Direct cell to cell to create more virioids

□ Prions

- ▣ *Proteinaceous Infectious Particles*
- ▣ Similar to proteins that normally exist, but with a different, contagious tertiary conformation
 - Causes conformation of normal proteins in the organism into prions at exponential rate
 - Associated with neurodegenerative diseases, or *Transmissible Spongiform Encephalopathies*, or *TSE's*
 - Mad Cow disease, a BSE, ND in cows that can pass to humans